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RESEARCH ARTICLE

Physics-Informed Data-Driven Modeling of HVAC Systems: A Systematic Analysis

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ABSTRACT Heating, ventilation, and air conditioning (HVAC) systems account for a significant portion of energy consumption in developed countries. Accurately identifying their dynamics is crucial for developing effective controllers. However, it is challenging due to the system's nonlinearities and variations across building types. Data-driven approaches have shown great promise in modeling the dynamics of HVAC systems, replacing traditional ordinary differential equations. However, relying solely on data can lead to poor generalization, particularly when the training data is limited or when unknown disturbances, such as weather conditions and occupant behavior, are present. Physics-informed machine learning (PIML) techniques have been developed for integrating physical principles into machine learning methods to improve the accuracy and data efficiency of modeling of dynamical systems. This paper investigates PIML techniques that incorporate three different physical properties: monotonicity, boundedness, and system structure. These models are benchmarked against physics-agnostic machine learning (PAML) approaches and the gray-box modeling technique to further highlight the performance of PIML models in terms of accuracy, robustness, and data efficiency for modeling HVAC systems from real data. Experimental data collected from a real-world HVAC system are used for systematically studying and analyzing thirteen gray-box, PAML, and PIML modeling techniques in different scenarios. Our results demonstrate that PIML models outperform PAML models in predicting the temperature dynamics of HVAC systems, especially when the training data is limited, unreliable, or noisy, with accurate and robust performance. Furthermore, we identify which physical properties are the most beneficial for enhancing machine learning performance for HVAC system identification.

INDEX TERMS HVAC systems, physics-informed machine learning, Gaussian process, neural network, gray-box modeling, sparse identification of nonlinear dynamics, data-driven modeling.

I. INTRODUCTION

With rapid technological advancements, the increasing use of machines and systems has significantly contributed to global energy consumption, exacerbating energy shortages. In developed nations, building systems represent the largest energy consumers, accounting for approximately 40% of total energy usage in 2024 [1]. Among them, heating, ventilation, and air conditioning (HVAC) systems constitute nearly 50% of a building's total energy consumption [2].

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Therefore, optimizing HVAC efficiency is crucial for achieving energy conservation and sustainability. However, in practice, common types of controllers (or regulators) used in HVAC systems, such as proportional-integral-derivative controllers [3] or on-off controllers [4], may not optimize energy consumption or guarantee a comfortable environment for occupants. Advanced controllers such as model predictive control have been developed for HVAC systems to address these issues [5], which can reduce energy consumption by 22% to 27% [6]. Nevertheless, achieving this potential requires accurately identifying the system models to facilitate effective control and optimization strategies.

Traditionally, HVAC system identification has relied on first-principles (physics-based modeling), which involves deriving governing equations based on thermodynamic and fluid dynamic principles [7]. These models provide deterministic representations that remain valid across diverse operating conditions, assuming an accurate system representation. However, real-world HVAC systems exhibit highly nonlinear dynamics with unobservable factors such as wall insulation variations, external airflow from windows and doors, and structural inconsistencies, making precise first-principles modeling challenging [8]. Due to these limitations, data-driven models have emerged as an alternative. Machine learning-based approaches have demonstrated high accuracy in predicting HVAC system behavior purely from measured data [9]. Nevertheless, data-driven methods suffer from poor generalization beyond their training range, rendering them unreliable in dynamic environments [10].

To overcome these challenges, physics-informed machine learning (PIML) has gained traction as a promising approach for system identification [11], [12], [13]. PIML integrates domain knowledge by incorporating physics-based constraints into data-driven models, thereby blending the advantages of physics-based and data-driven methods. Generally, there are three primary strategies for embedding physics into machine learning: *observational biases*, *learning biases*, and *inductive biases* [14], [15]. Among these, the inductive biases offer superior robustness and reliability due to hard constraints that strictly guarantee physical properties, making them suitable for system identification.

Various PIML methods have been proposed and implemented in the literature for HVAC systems, such as ensemble methods [16], the monotonicity-informed model [17], and the ODE-informed model [18]. Although multiple studies have applied PIML techniques for learning and controlling HVAC systems [19], [20], [21], [22], they primarily rely on soft-constraint learning biases. Additionally, the incorporated physics are typically based on the system's ODEs, which are difficult to determine for complex HVAC systems. These studies also do not account for general physical properties such as monotonicity, boundedness, or structural characteristics. Moreover, no PIML methods with these properties have been implemented or analyzed for real HVAC systems.

PIML approaches have been reviewed for system identification in building energy systems (BES) [23], [24]. These methods are typically categorized into four groups based on the integration of physics: physics-informed inputs, physics-informed loss functions, physics-informed architectures, and physics-informed ensembles within data-driven models. The reviewed literature highlights the advantages of using PIML in BES and discusses various applicable methods. In particular, [24] demonstrated how PIML can be employed in building performance simulation systems as base models, offering an alternative to traditional physics-based or purely data-driven simulation techniques. However, these review papers, including [23] and [24], primarily summarize

individual methodologies and do not offer a comprehensive comparison among different PIML techniques. There remains a lack of systematic studies that categorize and analyze the benefits of PIML approaches for real-world HVAC systems, examining not only predictive accuracy but also robustness and data efficiency, and encompassing comparisons across both physics-informed and physics-agnostic models involving different physical properties.

This study addresses these research gaps in the literature by providing a systematic comparison and analysis of PIML approaches for system identification of HVAC systems. Specifically, we analyze some physical properties of HVAC systems, implement PIML methods that incorporate these properties, which have not been previously explored in this context, and compare their performance against PAML models using real-world data. We investigate the indoor temperature model in HVAC systems because this model is directly related to the comfort conditions for occupants, as well as the components with several unknown disturbances, such as outdoor temperature [25]. Hence, the physical properties employed to fuse into the data-driven models are established by the thermal dynamics of the room temperature system.

The indoor temperature dynamics exhibit three common physical properties: bounded output constraints, model structure, and partial monotonicity with respect to the control input. These properties are embedded in machine learning models through inductive bias techniques, resulting in four distinct PIML models that are used in this paper: a structure-informed model, a model with bounded output constraints, a model enforcing monotonicity, and a hybrid model that combines both boundedness and monotonicity constraints. For a comprehensive comparison, these PIML approaches are benchmarked against seven purely data-driven models, referred to as physics-agnostic machine learning (PAML) models (i.e., non-physics-informed), and one gray-box model that serves as a physics-based baseline of the indoor room temperature model in HVAC systems.

The comparison is conducted using experimental data collected from a real building energy system of Northern Arizona University, United States of America. Two scenarios are created from the collected data: one with the original dataset and the other with a filtered dataset. The original dataset contains raw measurements that capture realistic operating conditions, including all disturbances present in the HVAC system. In contrast, the filtered dataset is obtained by removing most disturbances from the original data. Furthermore, each model is trained on various training data sizes to investigate how data availability affects model accuracy. We published our code and data at [26].

Our **main contributions** are highlighted as follows.

- **Categorization:** This paper surveys modeling methods for HVAC systems, based on underlying physical properties/principles. We categorize PIML models for HVAC systems according to three physical properties:

boundedness, monotonicity, and structural constraints. To the best of our knowledge, such a categorization has not been proposed in previous studies of PIML methods for HVAC systems.

- **Implementation:** We implemented four PIML techniques, that have not been previously investigated for HVAC systems, along with seven PAML techniques and one gray-box technique. All methods were applied to real-world data for modeling the room temperature dynamics of an HVAC system. In addition, multiple experimental scenarios were designed with varying dataset sizes and data qualities to evaluate model performance under different conditions.
- **Comprehensive analysis:** Based on these scenarios, we conduct comparative analyses among PIML, PAML, and gray-box modeling approaches. The results indicate that PIML models generally outperform PAML models on both filtered and original datasets, achieving higher and more robust accuracy in identifying HVAC system dynamics while demonstrating better data efficiency.

The remainder of this paper is organized as follows. Section II reviews the literature and discusses the selection of methods for HVAC system identification. Section III describes the gray-box model, a baseline method when comparing with machine learning techniques. Sections IV and V present the physics-agnostic and physics-informed machine-learning methods, respectively, that are implemented in the room temperature model. Section VI details the methodology, including data collection, preprocessing, and model setup. Section VII presents the experimental results and a comparative analysis of the thirteen models. Finally, Section VIII concludes the paper.

II. LANDSCAPE OF DATA-DRIVEN METHODS FOR MODELING HVAC SYSTEMS

A. HVAC SYSTEMS

This section introduces the dynamic model of an HVAC system, in which the room temperature is regulated by a fan-powered box (FPB). The thermal behavior of the room is affected by unmeasured disturbances such as return air and external air infiltration through doors and windows, as illustrated in Figure 1. The FPB consists of two primary components: the discharge supply temperature, regulated by a reheat coil, and the air mass flow rate, provided by the air handling unit (AHU). In this study, the FPB fan is treated as an unmodeled component, as we assume its operation is neither directly controllable nor measurable but by the thermostat. However, the fan introduces additional dynamics and significantly affects system behavior by directly influencing both the airflow velocity and the reheating process.

The temperature dynamics are governed by three measurable variables: the room temperature T_r , the discharge supply temperature T_s , and the air mass flow rate $\dot{m}_s$. The continuous-time behavior of the room temperature is

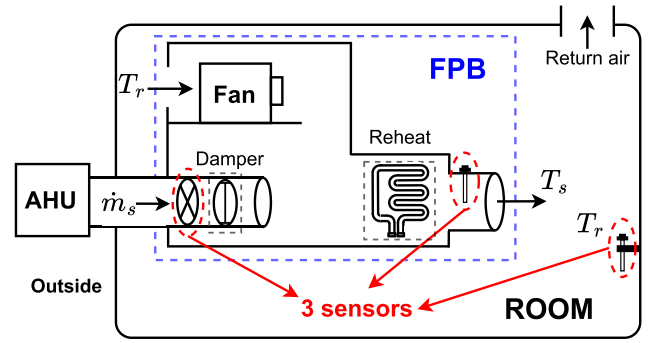


FIGURE 1. Indoor temperature model of an HVAC system. There are three sensors to measure and collect data.

described by

$$\frac{dT_r(t)}{dt} = f_{ct}(T_r(t), T_s(t), \dot{m}_s(t)). \quad (1)$$

Besides the continuous model, we adopt the discrete-time model of the HVAC system as follows

$$T_r^{k+1} = f_{dt}(T_r^k, T_s^k, \dot{m}_s^k), \quad (2)$$

where $T_r^k = T_r(t_k)$, $T_s^k = T_s(t_k)$, $\dot{m}_s^k = \dot{m}_s(t_k)$, and t_k is the k -th sampling instant. Higher-order discrete-time dynamics [27], which includes terms such as T_r^{k-1} , T_s^{k-1} , $\dot{m}_s^{k-1}$, ..., can be used to provide more accuracy in describing the dynamics of HVAC systems. In the context of the paper, we focus on a comprehensive comparison of data-driven modeling methods for HVAC systems. Thus, we use (2) as a common predictive model for data-driven modeling methods afterward, which use current state and current inputs for inferring future room temperature.

B. DATA-DRIVEN MODELING

This section classifies the data-driven models to be investigated in this paper into three groups: gray-box models, PAML models, and PIML models. Within each group, the literature review will be presented, and we will decide which methods to implement in HVAC systems.

1) GRAY-BOX MODELING

Gray-box models combine elements of both physics-based (first principles) and data-driven modeling approaches. The gray-box model is developed by defining the unknown parameters and optimizing them from the collected data. They incorporate partial physical knowledge of the system, such as fundamental equations or simplified relationships while relying on empirical data to estimate unknown parameters or capture unmodeled dynamics [28], [29]. This hybrid framework allows gray-box models to strike a balance between interpretability and flexibility, making them particularly suitable for complex engineering systems, where complete physical modeling may be impractical or computationally expensive.

In the context of HVAC systems, gray-box techniques often employ the resistance-capacitance (RC) model [8], which analogizes thermal behavior using electrical circuit representations. This approach captures the heat transfer characteristics of building components through resistance and capacitance elements, providing a compact yet interpretable structure. As an application of gray-box modeling, the RC model can offer higher prediction accuracy compared to purely physics-based models, which require manual measurement and calculation of parameters based on detailed knowledge of HVAC components such as wall insulation and window properties. A detailed formulation of the RC model, along with parameter estimation and inference methods, is presented in section (III).

2) PHYSICS-AGNOSTIC MACHINE LEARNING (PAML) MODELING

Data-driven models, which rely purely on data without a predefined model structure, have gained popularity for their ability to capture system behavior. By learning system behavior directly from data, a data-driven model can adapt to nonlinearities, uncertainties, and time-varying characteristics that are often challenging for traditional physics-based models. There are several ML methods used for data-driven models: linear regression (LR) [30], decision trees (DT) [39], artificial neural network (NN) [31], Gaussian process (GP) [32], and sparse identification of nonlinear dynamics (SINDy) [33]. However, their lack of physical interpretability and limited generalization, particularly in extrapolative scenarios, remain significant challenges [10].

The LR models the relationship between a dependent variable and one or more independent variables by fitting a linear equation to observed data. As a tree-based method, DT [39] and RF [40] are non-parametric machine learning techniques that use a hierarchical structure of decision rules to split data into subsets, enabling flexible modeling of complex, nonlinear relationships between features and target variables. GP [32], a probabilistic approach, provides uncertainty quantification, making it well-suited for system identification tasks. Besides that, SINDy [33] is a recent data-driven modeling approach that identifies the dynamical systems by selecting a small set of active terms from a large candidate library using sparse regression. However, since the SINDy technique relies solely on data for training without incorporating any physical constraints, we classify it under the PAML group.

NNs are computational models inspired by the structure of the human brain, consisting of interconnected layers of nodes (neurons) that learn to approximate complex functions from data [31]. NNs have been broadly applied for the identification of HVAC systems [41] and demonstrated strong performance in control [42]. As a variant of NN, residual neural networks (ResNet) [34] introduces the concept of residual connections that help mitigate the vanishing gradient problem. For dynamical systems, ResNet has been

employed to enforce smooth transitions in the learned system dynamics, thereby enhancing numerical stability and improving the reliability of simulations in continuous-time models [43]. Apart from ResNet, feed-forward NNs lack temporal awareness, limiting their ability to capture time-series patterns. To address this limitation, this study incorporates recurrent neural networks (RNNs) [36], which explicitly model time-dependent relationships. RNNs enable the model to make accurate multi-step predictions, ensuring that the predicted temperature evolution aligns with physical reality. With respect to smoothness and causality guarantees, neural ordinary differential equations (NODEs) [35], a class of models that represent system dynamics using a neural network to establish differential equations, are capable of capturing continuous-time behavior.

Besides ResNet, RNN, and NODE, deep operator network (DeepONet) [44], and Fourier neural operator (FNO) [45] have been proposed to identify dynamical systems effectively. Rather than learning the system dynamics equations, these methods aim to learn the operators that govern the system's evolution. While DeepONet and FNO have demonstrated strong performance in system identification tasks, they typically require large amounts of training data to achieve accurate results. Therefore, these models are excluded from the comparative analysis, as this study focuses on evaluating the impact of incorporating physical priors to enhance machine learning performance under limited data availability.

In this paper, the PAML models, including LR, NN, SINDy, ResNet, NODE, and RNN, are compared to PIML models, which are discussed in the following section.

3) PHYSICS-INFORMED MACHINE LEARNING (PIML) MODELING

There are several techniques to incorporate physical constraints into ML models. Hybrid machine learning integrates physical models such as ODEs with machine learning models to model dynamical systems more precisely [46], where the physical models guarantee the most essential structure of the systems, so that ML models only need to learn components unaccounted for by the physical models. Additionally, stability-informed machine learning [47] and convexity-informed machine learning [48] are proposed for learning and control tasks because they are significant properties with the potential to significantly improve the accuracy of system identification. Nevertheless, these approaches require deep understanding of the physical models in each type of building energy systems. Therefore, they require significant efforts from experts for different buildings.

This study categorizes the common properties of HVAC systems into three groups: boundary constraints, structural characteristics, and monotonicity. Since the room temperature naturally fluctuates within a bounded range, this boundary can be used to constrain the predicted outputs. The system operates within predefined safety and

TABLE 1. Classification of thirteen data-driven models considered in this study.

Category	Full Name	Short name	Description	Ref.
Gray-box	Resistance-capacitance	RC	Electrical analogy-based thermal model	[9]
Physics-agnostic machine learning (PAML)	Linear regression	LR	Models data with a linear equation	[30]
	Neural network	NN	Multi-layer feed forward neural network	[31]
	Gaussian process	GP	Probabilistic model for uncertainty-aware predictions	[32]
	Sparse identification of nonlinear dynamics	SINDy	Identifies sparse equations for dynamical systems	[33]
	Residual neural network	ResNet	Neural network using residual connections	[34]
	Neural ordinary differential equation	NODE	Learns continuous-time ODEs	[35]
	Recurrent neural network	RNN	Sequential model for time-series data	[36]
Physics-Informed Machine Learning (PIML)	Block-structured neural network	BSNN	Models system dynamics in block-structured form	[10]
	Bounded neural network	BNN	Enforces bound constraints on network outputs	[14]
	Min-Max neural network	MMNN	Enforces input-output monotonicity	[37]
	Unbounded lattice neural network	ULNN	Enforces monotonic conditions	[38]
	Bounded lattice neural network	BLNN	Enforces output bounded and monotonic constraints	[38]

operational limits, necessitating bounded constraints to prevent physically invalid predictions [49]. Regardless of safety and operational limits, conventional NN models may generate extreme or unrealistic outputs, leading to instability when deployed in control systems. This work considers a bounded NN model (BNN) for enforcing bounded constraints within NN. The BNN model guarantees that all outputs remain within operational limits, ensuring physically meaningful predictions.

In the context of a room-temperature model, the structural property refers to the ability to decompose the input into distinct components, rather than feeding all variables directly into a machine learning model. Block-structured neural networks (BSNN) [10] refers to architectures in which the network is explicitly organized into modular or hierarchical blocks, each responsible for modeling a specific subset of the system's behavior or data structure. These blocks may be designed based on domain knowledge, such as physical subsystems in a control application, or to enforce structural properties such as sparsity, locality, or decomposability.

Monotonicity is another essential property in HVAC system modeling, as certain input-output relationships exhibit strict monotonic behavior. For our case, an increase in current supply temperature $T_s(t)$ should not lead to a decrease in future room temperature $T_r(t)$ under normal conditions. This study investigates two approaches for enforcing monotonicity constraints in neural networks: the min-max neural network (MMNN) [38] and the lattice neural network (LNN) [37]. These methods can guarantee monotonicity by constraining the parameters related to monotonic inputs to always be positive or negative. Moreover, LNN from the Tensorflow library [50] includes the boundary constraints of the model. Hence, this study implements two versions of the LNN technique for HVAC systems: unbounded LNN (ULNN) and bounded LNN (BLNN).

In a nutshell, this paper compares thirteen machine learning techniques for modeling HVAC systems, as listed and categorized in Table 1.

C. LEARNING AND PREDICTION

To train the data-driven models of the HVAC system, the general form of loss functions is given by

$$\mathcal{L}(\mathbf{T}_r, \hat{\mathbf{T}}_r; \Theta) = \mathcal{L}_{\text{data}}(\mathbf{T}_r, \hat{\mathbf{T}}_r; \Theta) + \mathcal{L}_{\text{reg}}(\Theta),$$

where $\mathcal{L}_{\text{data}}(\mathbf{T}_r, \hat{\mathbf{T}}_r; \Theta)$ represents the error terms between the predicted and actual room temperatures over the training dataset, $\hat{\mathbf{T}}_r = \{\hat{T}_r^i\}_{i=1}^{N_{\text{train}}}$ stands for the predicted vector obtained by feeding the input sequences $\{T_r^i, T_s^i, \dot{m}_s^i\}_{i=1, \dots, N_{\text{train}}}$ into the model, $\mathbf{T}_r = \{T_r^i\}_{k=1}^{N_{\text{train}}}$ represents for target vector containing the corresponding measured future room temperatures, N_{train} is the total number of training samples, and $\mathcal{L}_{\text{reg}}(\Theta)$ denotes the regularization term used to reduce model complexity and mitigate overfitting, such as the L2-norm of the model parameters. In the study, $\mathcal{L}_{\text{data}}(\mathbf{T}_r, \hat{\mathbf{T}}_r; \Theta)$ is chosen as a mean squared error (MSE) in almost all the data-driven models except the log marginal likelihood function used by the GP. In addition, Θ is the vector of model parameters selected via the following optimization problem

$$\Theta^* = \arg \min_{\Theta} \mathcal{L}(\mathbf{T}_r, \hat{\mathbf{T}}_r; \Theta). \quad (3)$$

The majority of the ML methods introduced in this paper adopt the discrete-time dynamics in (2), resulting in the following prediction (inference)

$$T_r^{k+1} = \mathcal{M}_{\text{dt}}(T_r^k, T_s^k, \dot{m}_s^k). \quad (4)$$

Besides the discrete-time models, this paper considers the resistance-capacitance method and the neural ordinary differential equation (ODE) method, leading to continuous-time models. We define $\mathbf{x}^k = T_r^k$ and $\mathbf{u}^k = [T_s^k, \dot{m}_s^k]^\top$ as the state and input vectors, respectively, and denote $\mathbf{z}^k = [T_r^k, T_s^k, \dot{m}_s^k]^\top$. For the continuous model (1), an ODE solver is implemented to predict the future room temperature from the current state (or initial state) and current control inputs

$$T_r^{k+1} = T_r^k + \int_{t_k}^{t_{k+1}} \mathcal{M}_{\text{ct}}(T_r(t), T_s^k, \dot{m}_s^k) dt, \quad (5)$$

where the current inputs including $T_s(t)$ and $\dot{m}(t)$ are assumed unchanged from time step t_k to t_{k+1} since samplings of inputs are only available at sampling instants $t_1, t_2, \dots$.

The goals of data-driven modeling methods presented in this paper are to learn the continuous- or discrete-time dynamics from data $\{T_r^i, T_s^i, \dot{m}_s^i\}_{i=1, \dots, N_{train}}$ to accurately predict the room temperature in unobserved time.

III. RESISTANCE-CAPACITANCE MODEL

The dynamics of the in-room temperature model can be described by a basic RC model [8] as follows

$$C \frac{dT_r}{dt} = K(T_a - T_r(t)) + \dot{m}_s(t)c_a(T_s(t) - T_r(t)) + b, \quad (6)$$

where T_a is the ambient (outside) temperature, C represents the room thermal capacitance—ability to store heat, K denotes the thermal conductance—ability to conduct heat, c_a is the specific heat capacity of air—defining the energy required to change air temperature per unit mass, and b denotes bias for unmodeled heat gains or losses from the occupants or the return air inside the room.

It is a challenging task to find the parameters in equation (6) purely by physical principles, because of time-varying components of HVAC, such as the walls, pipeline, and air quality. In addition, there are numerous different types of buildings, so it would not be easy to define these parameters directly from the theoretical calculation of physics. For this reason, data-driven techniques are used to identify these parameters for a gray-box RC model from collected data.

We integrate the model (6) to calculate the predicted room temperature by the following

$$\begin{aligned} \hat{T}_r(t) &= RC(T_r^k, T_s^k, \dot{m}_s^k) \\ &= T_r^k + \frac{1}{C} \int_{t_k}^t [K(T_a - T_r(\tau)) \\ &\quad + \dot{m}_s(t_k)c_a(T_s(t_k) - T_r(\tau)) + b] d\tau. \end{aligned} \quad (7)$$

For the prediction of the RC model, $\hat{T}_r^{k+1} = \hat{T}_r(t_{k+1})$. From (7), we solve the parameters $\Theta_{RC} = [C, K, T_a, c_a, b]^\top$ by the following optimization problem (3). Due to the internal energy conservation from thermodynamics to get ODE, the RC model gives us a specific model that can guarantee generalization abilities, even when training with a small amount of data, for which we have only five parameters that need to be defined. However, there are many terms we did not know and did not include in the equation (6), so this model may not give us precise results when predicting if the unknown disturbances significantly affect the dynamics of HVAC systems.

IV. PAML FOR HVAC SYSTEMS

A. LINEAR REGRESSION

Linear regression (LR) is a classical machine learning method for system identification, particularly suited for linear systems [30], [51]. This approach describes the outputs of a

system as a linear combination of input features, assuming a direct linear relationship between the input variables and the system's response. In the context of HVAC modeling, LR predicts the temperature at the next time step as follows

$$\hat{T}_r^{k+1} = a_1 T_r^k + a_2 T_s^k + a_3 \dot{m}_s^k + a_4, \quad (8)$$

where $\Theta_{LR} = [a_1, a_2, a_3, a_4]^\top$ are the learnable parameters. These parameters are estimated by minimizing the MSE loss over the training dataset via the optimization problem as presented in (3). LR offers a simple and computationally efficient approach for HVAC system modeling, particularly when the training data is restricted. In the case where the system behavior deviates from linear interactions of the inputs, the LR model is adequate for capturing the dynamics accurately. However, for highly nonlinear HVAC systems or scenarios where input-output relationships exhibit complex dependencies, advanced modeling techniques such as neural networks or PIML may provide better accuracy of identification and generalization than the LR model.

B. SPARSE IDENTIFICATION OF NONLINEAR DYNAMICS

Sparse identification of nonlinear dynamics (SINDy) [33] is considered an improvement of the polynomial regression method. In that, it is not only limited by the use of polynomial terms, but we can also create our libraries to add the terms we need, such as square root, sine functions, and logarithmic functions in SINDy. Moreover, to avoid overfitting problems and utilize the sparsity of the dynamics, in the training process, it incorporates the L_1 -regularization term, also known as LASSO (least absolute shrinkage and selection operator) [52], to diminish terms that do not contribute to the prediction model and keep only the important terms from the entire libraries for the trained model.

We create a library of candidate functions $\beta(\mathbf{z}^k)$, with $\mathbf{z}^k = [T_r^k, T_s^k, \dot{m}_s^k]^\top$, where some components might represent the dynamics, constructed in (9). By following the RC model, this paper uses the polynomials for the library

$$\beta(\mathbf{z}^k) = [1, T_r^k, T_s^k, \dot{m}_s^k, (T_r^k)^2, T_r^k T_s^k, \dots]^\top. \quad (9)$$

The identification of the dynamics is formulated as a sparse regression problem. The objective is to find a sparse matrix of coefficients Θ_{SINDy} such that

$$\hat{T}_r^{k+1} = \beta(\mathbf{z}^k) \Theta_{SINDy}.$$

This can be achieved using sparse regression techniques such as LASSO or sequential thresholding

$$\Theta_{SINDy}^* = \arg \min \mathcal{L}_{data}(\mathbf{T}_r, \hat{\mathbf{T}}_r; \Theta_{SINDy}) + \lambda \|\Theta_{SINDy}\|_1,$$

where λ is a regularization parameter that promotes sparsity. SINDy offers several advantages, such as producing parsimonious models that are simple and interpretable by promoting sparsity, can handle a wide range of dynamical systems by choosing appropriate candidate functions, and effectively identifies the governing equations from limited data. However, there are practical considerations to keep

in mind: the choice of candidate functions $\beta(\mathbf{z})$ should be carefully made to balance model complexity and accuracy, and the regularization parameter λ needs to be tuned to achieve the desired sparsity. In addition, high-quality data are crucial for accurate model identification, especially when dealing with noisy measurements.

C. NEURAL NETWORKS

A well-known technique for approximating dynamical functions is neural networks (NNs) [31]. Due to the universal approximation theory [53], NNs can approximate any nonlinear function. Also, there is much research that employs NN in HVAC systems [9], [54], [55], and they have shown high accuracy for system identification tasks. NN includes three types of layers: the input layer, the hidden layer, and the output layer. Where the input layer receives the initial data for processing, the hidden layers are intermediate layers that perform computations and feature extraction, and the output layer produces the final prediction. Neurons or nodes in each layer are considered by nonlinear activation functions such as $\tanh()$ or $\text{ReLU}()$ in the hidden layers. The NN model with one hidden layer for the HVAC system is presented in (10)

$$\begin{aligned}\hat{T}_r^{k+1} &= \text{NN}(T_r^k, T_s^k, \dot{m}_s^k) \\ &= \text{NN}(\mathbf{z}^k) = \mathbf{W}_2 f_1(\mathbf{W}_1 \mathbf{z}^k + \mathbf{b}_1) + \mathbf{b}_2,\end{aligned}\quad (10)$$

where $\mathbf{W}_i$ and $\mathbf{b}_i$ ($i = 1, 2$) are the weights and biases of the NN model. The optimization problem to address the parameters of the NN (Θ_{NN}) is presented as follows (3).

NN can be adopted to estimate these dynamic equations with large and rich data for any nonlinear, complex dynamical systems. Nevertheless, this method can only guarantee its approximation ability in the training data. If the data is not in the range of the training set, NN may not generalize well.

In this study, a neural network architecture consisting of three fully connected layers is employed for the NN model and all NN-based models across the model categories.

D. GAUSSIAN PROCESS

Gaussian process (GP) is a nonparametric machine learning approach widely used to approximate unknown dynamical systems [56], [57], particularly building systems for verification [58], control [59], and fault detection [60]. The GP model defines a distribution over functions, enabling flexible adaptation to observed data. Given a training data set, GP constructs a distribution that provides both a mean estimate and an uncertainty measure, making it particularly useful for noisy systems. The use of GP model bases on the assumption that concerned variables follow the normal distribution.

Compared to other ML methods, GP has an advantage in dealing with noisy data. To be more specific, let us consider the observation model from discrete-time dynamics (2): $\tilde{T}_r^{k+1} = f_{\text{dt}}(\mathbf{z}^k) + \mathcal{N}(0, \sigma_n^2)$ where $\mathbf{z}^k = [T_r^k, T_s^k, \dot{m}_s^k]^\top$, noisy observation $\tilde{T}_r^{k+1}$, and noise variance σ_n^2 . Here, we assume that the latent function $f_{\text{dt}}(\mathbf{z}) \sim \mathcal{N}(\mu(\mathbf{z}), \kappa(\mathbf{z}, \mathbf{z}'))$ where $\mu(\mathbf{z})$

represents given (prior) mean function and $\kappa(\mathbf{z}, \mathbf{z}')$ is a given kernel function. The covariance function $\kappa(\mathbf{z}, \mathbf{z}')$ governs the smoothness and structure of the function. A common choice is the radial basis function [32]. Denote $\mathbf{Z} = \{\mathbf{z}^k\}_{k=1, \dots, N_{\text{train}}}$ and $\mathbf{y} = \{\tilde{T}_r^{k+1}\}_{k=1, \dots, N_{\text{train}}}$ as a vector of inputs and vector of observation (noisy data). The joint distribution of the observed data $\mathbf{y}$ and tested output T_r^{k+1} is given by the following multivariate distribution

$$\begin{bmatrix} \mathbf{y} \\ T_r^{k+1} \end{bmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{bmatrix} \mathbf{K}_\sigma(\mathbf{Z}, \mathbf{Z}) & \mathbf{K}(\mathbf{Z}, \mathbf{z}^k) \\ \mathbf{K}(\mathbf{z}^k, \mathbf{Z}) & \kappa(\mathbf{z}^k, \mathbf{z}^k) \end{bmatrix}\right),$$

where $\mathbf{K}_\sigma(\mathbf{Z}, \mathbf{Z}) = \mathbf{K}(\mathbf{Z}, \mathbf{Z}) + \sigma_n^2 \mathbf{I}$, $\mathbf{K}(\mathbf{z}^k, \mathbf{Z}) = \mathbf{K}(\mathbf{Z}, \mathbf{z}^k)^\top = [\kappa(\mathbf{z}^k, \mathbf{z}^1) \dots \kappa(\mathbf{z}^k, \mathbf{z}^{N_{\text{train}}})]$, and $\mathbf{K}(\mathbf{Z}, \mathbf{Z})$ is the covariance matrix with elements $K_{ij} = \kappa(\mathbf{z}^i, \mathbf{z}^j)$. The posterior predictive distribution for function values at test inputs $\mathbf{z}^k$ is

$$\begin{aligned}T_r^{k+1} | \mathbf{y} &\sim \mathcal{N}(\hat{\mu}^{k+1}, \Sigma^{k+1}), \\ \hat{\mu}^{k+1} &= \mu(\mathbf{z}^k) + \kappa(\mathbf{z}^k, \mathbf{Z}) \mathbf{K}_\sigma^{-1}(\mathbf{Z}, \mathbf{Z}) \mathbf{y}, \\ \Sigma^{k+1} &= \kappa(\mathbf{z}^k, \mathbf{z}^k) - \mathbf{K}(\mathbf{z}^k, \mathbf{Z}) \mathbf{K}_\sigma^{-1}(\mathbf{Z}, \mathbf{Z}) \mathbf{K}(\mathbf{Z}, \mathbf{z}^k).\end{aligned}$$

The kernel's hyperparameters, such as the length scales, amplitudes of the kernel, and noise variance σ_n^2 , are optimized by maximizing the log marginal likelihood, given by

$$\begin{aligned}\log p(\mathbf{y} | \mathbf{Z}) &= -\frac{1}{2} \mathbf{y}^\top \mathbf{K}_\sigma^{-1}(\mathbf{Z}, \mathbf{Z}) \mathbf{y} \\ &\quad - \frac{1}{2} \log \det \mathbf{K}_\sigma(\mathbf{Z}, \mathbf{Z}) - \frac{n}{2} \log(2\pi),\end{aligned}\quad (11)$$

where n denotes the number of data points (the dimensionality of the observation vector $\mathbf{y}$). For prediction, we choose $\hat{T}_r^{k+1} = \hat{\mu}^{k+1}$. GP regression provides probabilistic predictions, offering both a mean estimate and uncertainty quantification, which is crucial to assessing the confidence of the model in HVAC control applications. However, GP faces computational challenges due to the need to invert the covariance matrix, making it impractical for large data sets. This computational complexity scales significantly, increasing the training time for high-dimensional data.

E. RESIDUAL NEURAL NETWORKS

Residual neural networks (ResNet) had been proposed in [34] to improve deep neural network models, avoiding vanishing gradient issues when training, and applied to HVAC systems in [61], [62], and [63]. In the case study of training with a small amount of data and using the narrow neural network model, avoiding the vanishing gradient issue may not improve the accuracy of predicting systems. However, in terms of the control aspect, ResNet can be analyzed as a numerical solution for the continuous model, which is an Euler approximation [64], that keeps the current information of the system, which contributes directly to the output prediction. Equation (12) shows how to implement ResNet for HVAC systems, where we keep the current room temperature as a skip connection component in the ResNet

technique

$$\begin{aligned}\hat{T}_r^{k+1} &= \text{ResNet}(T_r^k, T_s^k, \dot{m}_s^k) \\ &= T_r^k + \text{NN}_{\text{Res}}(T_r^k, T_s^k, \dot{m}_s^k).\end{aligned}\quad (12)$$

Similar to NN, we train the ResNet model based on the optimization problem (3) to get the parameters Θ_{Res} , which includes the weights and biases of the ResNet model. The advantage of ResNet is that it keeps information on the current temperature and approximates the difference between two steps of room temperature instead of learning the future temperature directly. In small sampling time dynamics, ResNet will reduce the regions of predictions because ResNet just considers the changes between current and future temperatures. In contrast, it could be a weak point if we have a large time-step size where the temperature will change significantly, and it has a large region of predictions.

F. NEURAL ORDINARY DIFFERENTIAL EQUATION

The neural ordinary differential equation (NODE) is an improvement of the ResNet model. NODE has been promoted in [35], and it illustrated auspicious results in identifying regression tasks. The NODE technique is also implemented in HVAC systems for learning and control tasks in various studies [65], [66], [67]. In the previous section, ResNet is considered an Euler numerical approximation method. On the other hand, NODE is more general in using the integral of a neural network instead of adding a neural network directly as ResNet. They consider the continuous part in dynamical systems instead of only discrete models. To utilize the NODE technique for HVAC systems, we determine the structure of NODE as performed in equation (13) for one-step prediction

$$\begin{aligned}\hat{T}_r^{k+1} &= \text{NODE}(T_r^k, T_s^k, \dot{m}_s^k) \\ &= \hat{T}_r^k + \int_{t_k}^{t_{k+1}} \text{NN}_{\text{NODE}}(T_r(t), T_s^k, \dot{m}_s^k) dt.\end{aligned}\quad (13)$$

An important part of equation (13) is calculating the integral using collected data. A numerical integration method, such as a Runge–Kutta method [68], is used for computing $\hat{T}_r^{k+1}$ from the initial value T_r^k , T_s^k , and $\dot{m}_s^k$. The parameter Θ_{NODE} are optimized by the optimization problem (3).

When using NODE in HVAC systems to learn continuous models, we expect this model to outperform ResNet in accuracy, as NODE is a more effective solution for computing integrals than the Euler method. However, for this method, the computational cost will be high due to the integral part. Moreover, during integral calculation, NODE assumes the control inputs will not change and considers the function as a continuous model. If the time-step size is large, it results in a significant accumulation of errors for the integration.

G. RECURRENT NEURAL NETWORKS

Recurrent neural network (RNN) [36] has been applied to various applications, including HVAC modelling [69], energy consumption forecasting [70], and control [71]. This

method was used to obtain the outputs of the previous hidden layer as inputs for the next hidden layer. Consequently, the information of all hidden layers will be kept during the training process. In this study, we apply RNN for modeling the HVAC system, aiming to improve its accuracy and robustness in multistep predictions, where the accuracy is maintained for multiple prediction steps. Correctly predicting multiple future steps is essential for model predictive control [72].

In this paper, we utilize an RNN model for multiple-step predictions, employing different training methods. First, we define the multiple predictions in training data as follows

$$\begin{cases} \hat{T}_r^{k+1} = \text{RNN}(T_r^k, T_s^k, \dot{m}_s^k), \\ \hat{T}_r^{k+2} = \text{RNN}(\hat{T}_r^{k+1}, T_s^{k+1}, \dot{m}_s^{k+1}), \\ \vdots \\ \hat{T}_r^{k+n+1} = \text{RNN}(\hat{T}_r^{k+n}, T_s^{k+n}, \dot{m}_s^{k+n}), \end{cases}\quad (14)$$

where RNN stands for an NN model with three inputs and one output, n is the number of steps we want to predict in the future. On the other hand, the architecture of RNN here is the same as that of NN, but with a different training technique. In addition, the control inputs T_s^{k+i} and $\dot{m}_s^{k+i}$ are used from the real data in the training data set. This process continues until the predicted T_r^{k+n} is obtained.

We define the loss function to train our RNN model in equation (15), and equation (16) shows how we detect the parameters Θ_{RNN} of the RNN model

$$\mathcal{L}_{\text{RNN}} = \sum_i \lambda_i \mathcal{L}_i, \quad \mathcal{L}_i = \mathcal{L}(\mathbf{T}_r, \hat{\mathbf{T}}_r; \Theta_{\text{RNN}}), \quad (15)$$

$$\Theta_{\text{RNN}}^* = \arg \min_{\Theta_{\text{RNN}}} \mathcal{L}_{\text{RNN}}, \quad (16)$$

where λ_i is the weight that defines which term we want to prioritize. The RNN approach will be beneficial for optimal controllers that require predicting multiple future values, and this method could maintain the accuracy of various predictions. However, it may not be able to accurately approximate one-step ahead predictions when the training data is large enough for the NN model.

V. PIML FOR HVAC SYSTEMS

This section presents four PIML methods that incorporate three general types of physical properties in HVAC systems, including structural, monotonic, and boundedness constraints.

A. BLOCK-STRUCTURED NEURAL NETWORKS

The block-structured neural network (BSNN) for HVAC is inspired by the neural network model used for the linear model of HVAC systems [10]. They use two different NNs for describing the independent impacts of $\mathbf{x}$ and $\mathbf{u}$ as follows

$$\mathbf{x}^{k+1} = \text{NN}_1(\mathbf{x}^k) + \text{NN}_2(\mathbf{u}^k), \quad (17)$$

where NN_1 and NN_2 are the forward NNs used to capture the dynamics of HVAC. As can be seen from the RC model

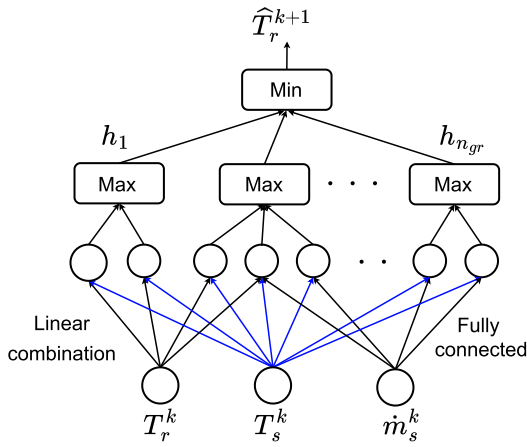


FIGURE 2. Min-max neural network (MMNN) architecture.

in (6), there are cross impacts of input and state, that is, an interconnection between the state (room temperature) and the inputs (mass flow rate and supply discharge temperature). Therefore, the separated input-state model (17) is likely conservative for modeling HVAC systems. Regarding physical principles for thermal dynamics, we consider convection, which is a nonlinear part that depends only on the room temperature. Moreover, the RC model did not illustrate the effect of the unknown disturbance fan (6), which gets the T_r as the input and proceeds to the FPB box and the future room temperature. We define the BSNN model as follows

$$\begin{aligned}\hat{T}_r^{k+1} &= \text{BSNN}(T_r^k, T_s^k, \dot{m}_s^k) \\ &= \text{NN}_1(T_r^k) + \text{NN}_2(\text{NN}_1(T_r^k), T_r^k, T_s^k, \dot{m}_s^k),\end{aligned}\quad (18)$$

where the NN_1 represents the dynamics of the fan, and NN_2 represents the general system. Similar to other neural network models, we also use the mean square error loss function to find the optimal values for the BSNN model. The optimization issue finds the parameters Θ_{BSNN} via (3).

By decoupling the state and control dynamics and leveraging the function approximation capabilities of neural networks, it is possible to achieve more accurate predictions and better control strategies for systems exhibiting nonlinear behavior. However, when dealing with linear or simple systems, BSNN may not be effective, particularly with small training data that BSNN may model noises, leading to overfitting.

B. MIN-MAX NEURAL NETWORKS

Min-max neural network (MMNN) [38] incorporates monotonicity into a neural network model. This algorithm has been demonstrated to be able to approximate any function by universal approximation theory [73]. MMNN architecture consists of only three layers: the input, max, and min layers (output constraints). Moreover, instead of using popular nonlinear activation functions, MMNN employs Min and Max functions. In addition, dealing with inputs that increase

monotonically with output, their weights in the linear combination are defined as positive values, such as the blue line in Figure 2, which describes the architecture of MMNN. The mathematical model is illustrated in

$$\begin{aligned}\hat{T}_r^{k+1} &= \min_i(h_i), \\ h_i(\mathbf{z}) &= \max(\mathbf{W}_i \mathbf{z}^k + b_i),\end{aligned}\quad (19)$$

where $\mathbf{W}_i$ and b_i are the weights and biases of linear combinations for each group $i = [1, 2, \dots, n_{gr}]$. The weights and biases $\Theta_{\text{MMNN}} = \{\mathbf{W}_i, b_i\}_{i=1, \dots, n_{gr}}$ are determined via (3).

MMNN provides efficient constraints for modeling monotonic functions and can be applied to any nonlinear system based on the universal approximation theory. However, it lacks the representational advantages of deep neural networks [74], as it typically consists of only three layers. Therefore, to model complex systems, MMNN may require a large network width to achieve sufficient expressiveness. Moreover, there is no specific method for grouping the linear combination layers for the minimum and maximum functions, which can make it difficult to obtain optimal parameters during training.

C. BOUNDED NEURAL NETWORKS

In practice, all dynamical systems have their safe range conditions. This is also an essential condition of HVAC systems. Especially considering temperature, to guarantee a comfortable room temperature, the range should be 68 to 77 degrees Fahrenheit. Due to this property, ML models for HVAC systems could implement a constrained function for their output (which is the room temperature T_r in our case). Here, we will constrain T_r to the range from the lowest temperature T_l to the highest temperature T_h . We define the predictions of the bounded neural network (BNN) model as follows

$$\begin{aligned}\hat{T}_r^{k+1} &= \text{NN}(T_r^k, T_s^k, \dot{m}_s^k), \\ \hat{T}_r^{k+1} &= \text{ReLU}(T_h - T_l - \text{ReLU}(T_h - \hat{T}_r^{k+1})) + T_l.\end{aligned}\quad (20)$$

From (20), the output of machine learning models will be trimmed if it goes over the boundary $[T_l, T_h]$. For this reason, the safe region of output is always secured. The optimization problem (3) illustrates how we train the parameters Θ_{BNN} for the BNN model.

BNN with an inductive bias for boundedness constraints will always guarantee that the output prediction falls within the system's suitable range, thereby improving accuracy even with a small training data set. However, the constraints could be conservative for the training process if we do not accurately determine the output limits.

D. DEEP LATTICE NEURAL NETWORKS

Another approach for enforcing monotonicity in neural networks is lattice neural networks (LNN) [37], which consists of three primary layers: a calibration layer, a linear embedding layer, and a lattice layer. The architecture of a

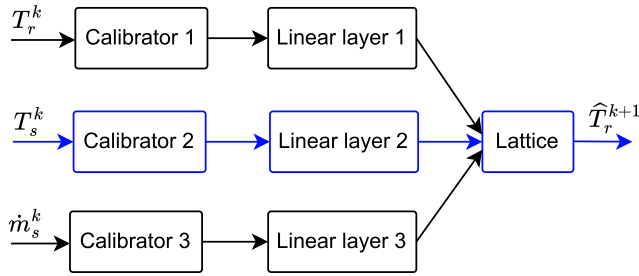


FIGURE 3. Lattice neural networks (LNN) architecture.

lattice network is illustrated in Figure 3. Lattice models are interpolated look-up tables, providing an efficient class of nonlinear functions that can be constrained to be monotonic by enforcing sparse linear inequalities on the parameters (calibrator). The output is obtained by aggregating multiple lattice functions in an ensemble

$$\hat{T}_r^{k+1} = \frac{1}{G} \sum_{g=1}^G y_g,$$

where G represents the number of lattices in the ensemble, and y_g denotes the output of the g -th lattice function. Each lattice function is a linearly interpolated multidimensional look-up table

$$y_g = \sum_{v \in \mathcal{V}} w_v \prod_{i=1}^3 l_i(v_i),$$

where $\mathcal{V}$ is the set of vertices (grid points) in the lattice space, w_v are the learned parameters at each vertex, and $l_i(v_i)$ is the linear interpolation function for dimension i , ensuring smooth transitions between grid points. To ensure the monotonicity of T_r^{k+1} with respect to T_s^k , the following constraint is enforced on the learned parameters: whenever two lattice vertices differ only in the monotonic input dimension, the vertex with the larger input value must not have a smaller parameter value than the vertex with the smaller input value. The model parameters Θ_{LNN} are optimized via equation (3).

By incorporating monotonicity constraints, the LNN model ensures physically consistent predictions, particularly in control applications where the response of the system should be predictable with respect to variations in control inputs. However, despite its advantages in interpretability and structure, the LNN has a more complex model structure and higher computational cost compared to standard neural networks. This trade-off makes it less efficient for large-scale applications and high-dimensional data.

On the other hand, this study also aims to incorporate additional physical properties into machine learning models and investigate the performance improvements achieved by combining multiple physics-informed constraints, rather than relying on a single property. Specifically, for the LNN model, which is already informed by the monotonicity condition, we introduce an additional boundedness constraint. As a result, we consider two LNN variants: unbounded lattice

neural network (ULNN), which enforces monotonicity but remains unbounded, and bounded lattice neural network (BLNN), which incorporates both monotonicity and boundedness.

VI. METHODOLOGY

A. HVAC DATA COLLECTION

Description of the system: The room in the HVAC system analyzed in this study is located in a corner room on the first floor of the School of Informatics, Computing, and Cyber Systems at Northern Arizona University. As mentioned in Section I, this study will focus on learning the dynamics of the indoor temperature system within the entire HVAC system. The room is a laboratory used by researchers, equipped with personal computers and other electronic devices. The room has a transparent glass wall directly exposed to the outdoor environment, making it highly susceptible to solar radiation in the morning and to outdoor temperature fluctuations. Consequently, outdoor temperature fluctuations significantly influence indoor thermal dynamics.

The data were collected using the Alerton Building Management System (BACnet) from three sensors measuring the mass flow rate, supply temperature, and room temperature. We collected the data over six consecutive days in August, capturing thermal variations during the summer in Flagstaff, Arizona, USA. Measurements were recorded at 15-minute intervals, totaling 560 data points. The considered system is an FPD system as presented in Section II-A.

Description of the data scenarios: To comprehensively examine PIML compared to PAML techniques, we create two different data sets from the original data. They include a filtered data set, where the disturbance from the fan is filtered out, and the original data set with all the disturbances present. The filtered data set represents an ideal scenario with clean data, no disturbances, thus the data accurately represents the system dynamics, well aligned with the underlying physical equations. On the other hand, the unfiltered (original) data set represents a realistic scenario with contaminated data and unknown disturbances, with a complex impact of disturbances on the system dynamics, unmodeled by the physical equations. All the data are published in [26].

We train all data-driven models on different numbers of training sizes from 20 to 300 points for the original data and from 20 to 180 points for the filtered data, in 10-point increments. This setup enables an evaluation of how training data size influences the performance of physics-informed and physics-agnostic machine learning models. The models with good performance can guarantee robust results of high accuracy across various training data cases, ranging from small to large data sizes.

B. TRAINING-TESTING SET-UP

This subsection details the implementation of the data-driven models investigated in this study.

Data pre-processing: For all data-driven models, we use min-max normalization on the original data, except for the RC model (to preserve its physics structure). For example, for the room-temperature data, we normalize their values in Fahrenheit (F) to the range 0 to 1. Otherwise, for the predicted outputs, we will denormalize the range 0 to 1 to the original scale and then evaluate their performance on the test data sets for each scenario.

Implementation of models: For LR, RC, SINDy, and GP models, we use established libraries to ensure consistency and reproducibility. Implementation of LR is carried out by Linear Regression package from `scikit-learn` [75], and the RC model is identified by fitting a first-order nonlinear model to the data using `curve_fit` from `scipy.optimize` [76]. SINDy is implemented by `PySINDy` package [77] with the polynomial library fixed to degree 3 across all experiments. The parameter of L_1 regularization needs to be determined. In our study, we use cross-validation for training data with 10 folds to choose this parameter for SINDy. GP is implemented with `GPYtorch` [78], employing an RBF kernel with three different length scales for three inputs. We train GPs with different initializations, and hyperparameters are optimized through marginal log-likelihood maximization. Also, all the neural-network-based models except for Lattice NN are built by using `Pytorch` [79], and Lattice NN is developed by using `TensorFlow` [50].

NN implementation: For neural network-based models, several techniques are used to mitigate overfitting, particularly when the training data size is limited. We perform 10-fold cross-validation with different learning rates. The L_2 weight regularization is added to the loss function to discourage large weights. All NN models, including PAML and PIML methods, are initialized with the same structures such as the same number of neurons, weights, activation functions, and topologies. Initial weights are selected by the Xavier initialization method [80]. We employ step decay for the learning rate with a fixed decay rate. The learning rate varies from 0.01 to 0.4 in increments of 0.01, and the best learning rate is selected based on the minimum average loss across all validation folds. The best learning rate is the initial value and will be applied to the weight decay over the training epoch. Once the best learning rate is identified, each model is trained on the complete data set.

Testing metric: We select a test data set in which the outdoor temperature range differs significantly from that of the training data set. This selection enables a more comprehensive evaluation of the ML model's generalization capability. The prediction performance of each model is evaluated by the root mean square error (RMSE)

$$\text{RMSE} = \sqrt{\frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} (T_r^i - \hat{T}_r^i)^2},$$

where a smaller RMSE indicates a more accurate model. Moreover, we examine the effect of training data size on prediction accuracy by progressively increasing the number of training data points, and investigate how the performance of PIML models improves as the training data set grows.

VII. EXPERIMENTAL RESULTS

A. CASE STUDY: FILTERED DATA

Training and test data sets:

After filtering the original data, the total number of collected data was reduced from 560 to 230. In the filtered data set, time intervals between two consecutive data points are not identical, so we will not consider the RNN method in this case. Consequently, **twelve models** in Table 1 will be considered in this case, except for RNN. The filtered data set consists of points from 180 to 200, with a total of 20 points in the test set. The models are trained on training sets of varying sizes, ranging from 20 to 180 points in increments of 10 consecutive points.

General comparison: Table 2 shows the RMSE of twelve models in test data size, trained with data sizes from 20 to 180 in increments of 10 consecutive points. The RC model demonstrates high accuracy in identifying room temperature, achieving the smallest RMSE in most cases when trained with 17 different training data sizes. The findings indicate that a properly structured gray-box model, which integrates physical equations with empirical data, provides an accurate representation of the system dynamics. By contrast, models such as SINDy and unconstrained neural networks yield higher errors and greater variability, underscoring their shortcomings for presenting system dynamics in the absence of explicit physical constraints.

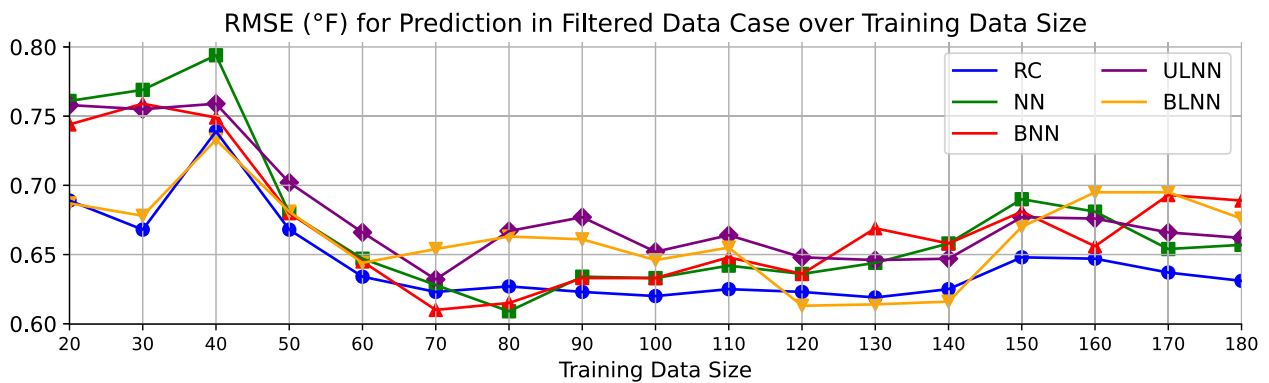
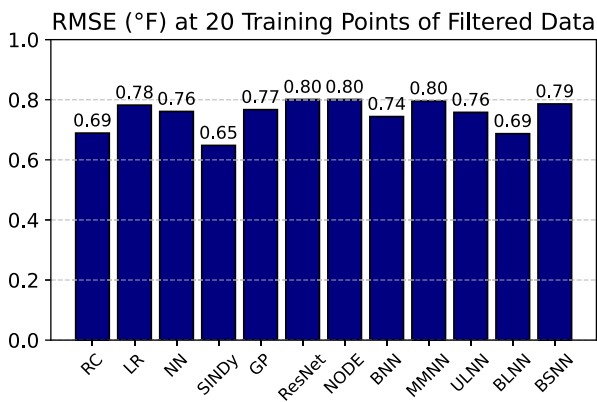
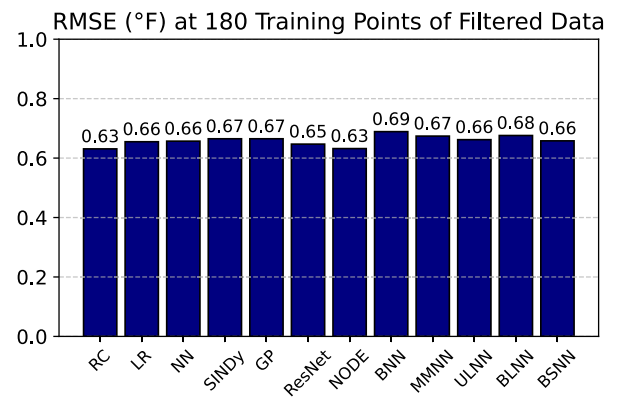
Figure 4 presents the RMSE of five model predictions, including RC, NN, BNN, ULNN, and BLNN, across different training data sizes. The figure shows that model accuracy improves gradually as the size of the training data increases. For example, the RMSE of the NN decreases from 0.761°F with 20 training data points to 0.657°F with 180 training data points. Moreover, the results highlight the robustness and stability of the RC model in the filtered data case. However, prediction accuracy can be ensured by a simple model like LR, since LR yields RMSEs that are competitive with those of NN, GP, and other machine learning models. Consequently, PIML models may not fully demonstrate their advantages in this scenario.

Comparison of PAML and PIML models: When using filtered data, PIML and PAML models do not show significant differences in terms of precision for HVAC system identification, as reported in Table 2.

From Figure 5, RC, SINDy, and BLNN outperform other models with the RMSE of 0.69°F, 0.65°F, and 0.69°F, respectively. SINDy performs well in this case because it can capture nonlinear components from its library

TABLE 2. RMSE ($^{\circ}\text{F}$) values of prediction across training data sizes in the filtered data case of twelve models.

	Gray-box	Physics-agnostic ML						Physics-informed ML				
Size	RC	LR	NN	SINDy	GP	ResNet	NODE	BNN	MMNN	ULNN	BLNN	BSNN
20	0.689	0.782	0.761	0.648	0.767	0.801	0.801	0.744	0.796	0.758	0.687	0.786
30	0.668	0.756	0.769	0.660	0.697	0.809	0.773	0.759	0.769	0.755	0.678	0.748
40	0.739	0.736	0.794	0.773	0.723	0.866	0.859	0.749	0.758	0.759	0.733	0.775
50	0.668	0.669	0.680	0.676	0.662	0.838	0.681	0.680	0.686	0.702	0.681	0.684
60	0.634	0.636	0.647	0.636	0.635	0.799	0.647	0.645	0.665	0.666	0.644	0.647
70	0.623	0.623	0.628	0.633	0.624	0.803	0.633	0.610	0.634	0.632	0.654	0.635
80	0.627	0.624	0.609	0.634	0.624	0.634	0.636	0.615	0.635	0.667	0.663	0.635
90	0.623	0.622	0.634	0.628	0.626	0.655	0.636	0.633	0.633	0.677	0.661	0.633
100	0.620	0.622	0.633	0.623	0.626	0.635	0.636	0.633	0.632	0.652	0.646	0.633
110	0.625	0.630	0.642	0.625	0.637	0.642	0.641	0.648	0.641	0.664	0.655	0.642
120	0.623	0.624	0.636	0.638	0.640	0.674	0.635	0.636	0.635	0.648	0.613	0.636
130	0.619	0.633	0.644	0.676	0.65	0.685	0.644	0.669	0.650	0.646	0.614	0.645
140	0.625	0.641	0.658	0.640	0.659	0.693	0.652	0.658	0.658	0.647	0.616	0.658
150	0.648	0.667	0.690	0.671	0.674	0.692	0.680	0.681	0.689	0.677	0.670	0.668
160	0.647	0.670	0.681	0.675	0.681	0.697	0.655	0.656	0.690	0.676	0.695	0.669
170	0.637	0.664	0.654	0.722	0.675	0.678	0.638	0.693	0.684	0.666	0.695	0.662
180	0.631	0.655	0.657	0.665	0.665	0.647	0.632	0.689	0.674	0.662	0.676	0.658

**FIGURE 4.** RMSE ($^{\circ}\text{F}$) values of prediction across training data cases (from 20 to 180 points) in the filtered data case.**FIGURE 5.** RMSE at 20 training points on filtered data.**FIGURE 6.** RMSE at 180 training points on filtered data.

during training. Furthermore, at 180 training data points in Figure (6), all models yield similar results, with the most significant difference being about 0.06°F , compared to 0.11°F at 20 training data points, representing nearly a

50% reduction in distinction. These results for the filtered case indicate that *when noises and disturbances are small or absent, PIML models do not demonstrate clear advantages in improving the accuracy of room temperature identification in*

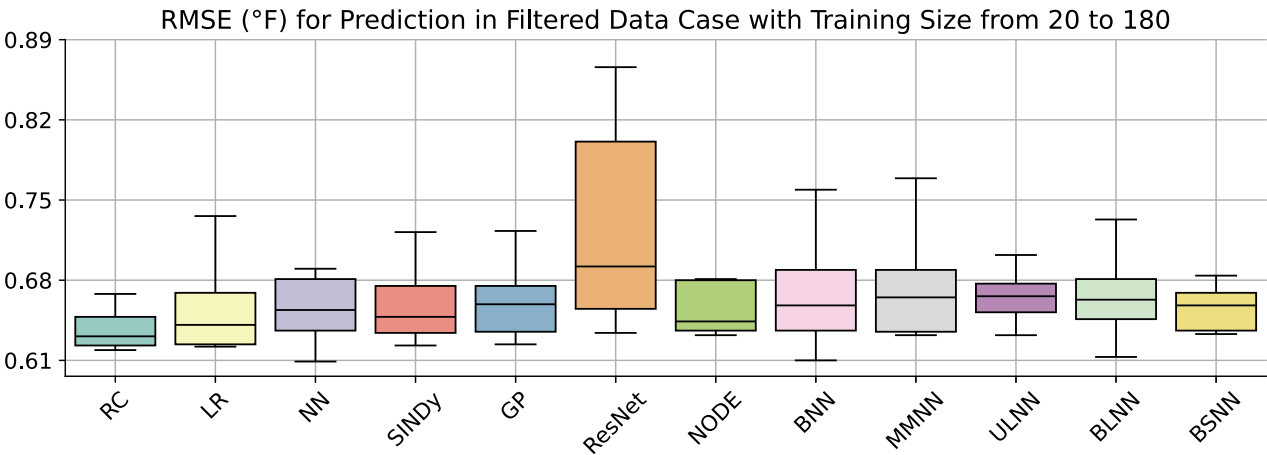


FIGURE 7. Box plot of RMSE (°F) for prediction performance across twelve models in the filtered data case, evaluated with training sizes ranging from 20 to 180. Lower medians show higher accuracy, while narrower ranges reflect greater robustness.

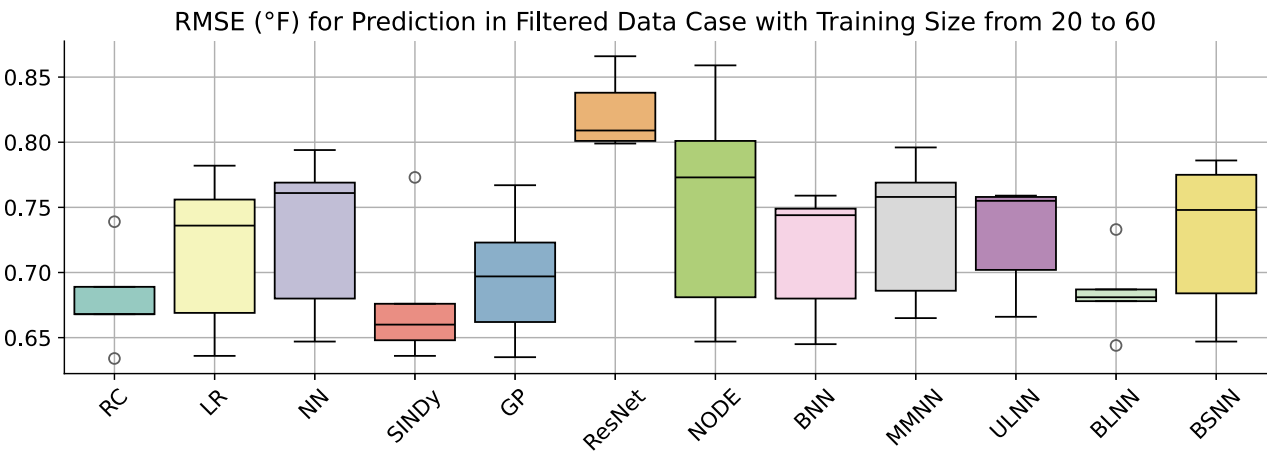


FIGURE 8. Box plot of RMSE (°F) for prediction performance across twelve models in the filtered data case, evaluated with training sizes ranging from 20 to 60. Lower medians show higher accuracy, while narrower ranges reflect greater robustness.

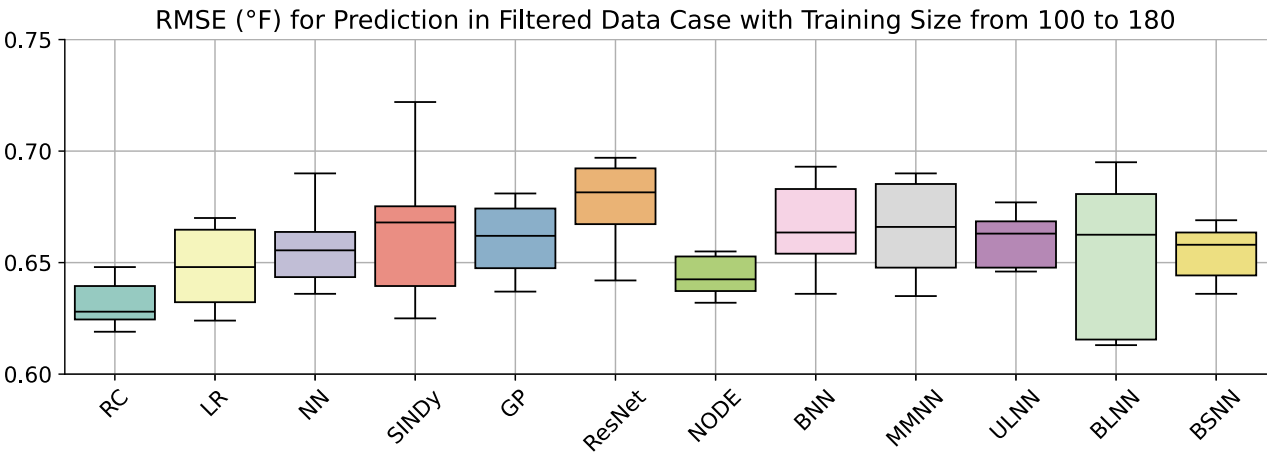


FIGURE 9. Box plot of RMSE (°F) for prediction performance across twelve models in the filtered data case, evaluated with training sizes ranging from 100 to 180. Lower medians show higher accuracy, while narrower ranges reflect greater robustness.

HVAC systems. Overall, the NN model outperforms the GP model in the filtered data case, as evidenced by consistently

lower RMSE in most scenarios. In conclusion, for the filtered data case, PAML and PIML have similar performance in

modeling HVAC systems, and PIML methods do not exhibit any substantial advantage.

Statistical comparison: Figures 7, 8, and 9 illustrate the RMSE predictions on the filtered data case across three training size intervals: (i) 20 to 180 data points as in Figure 7, (ii) 20 to 60 data points as in Figure 8, and (iii) 100 to 180 data points as in Figure 9. In that regard, (i) depicts the overall performance of the models for all training data sizes, (ii) demonstrates the performance of models with small training data sizes, and (iii) shows the performance for large training data sizes. Each box plot conveys both predictive accuracy (measured by median RMSE) and model robustness (indicated by the spread) across various data size scenarios.

- **Accuracy:** Across all three plots, the gray-box RC model consistently achieves the lowest or nearly lowest median RMSE, highlighting its superior performance, grounded in physical equations and trained on small amounts of collected data. For the smallest training data size, ranging from 20 to 60 points, Figure 8 presents three methods: the RC, BLNN, and SINDy models, which are the top performers, exhibiting the lowest medians, with RMSEs smaller than 0.68°F . In this context, PIML models are generally more accurate than PAML models (except for SINDy) with lower RMSEs. Notably, all PIML models outperform the NN model, with RMSEs lower than 0.75°F . On the other hand, with a larger training data set (from 100 to 180 data points), Figure 9 shows that the PAML models achieve performance similar to that of the PIML models. Indeed, the most significant difference in median RMSEs across twelve models is approximately 0.05°F . Still, the RC model's median RMSE is lower than that of other models. Overall, Figure 7 illustrates that the RC model is the most accurate with the smallest median RMSE of 0.63°F over 17 cases, as shown in Figure 7.
- **Robustness:** The width of each box plot, quantified by its interquartile range (IQR) or spread, indicates the robustness of the ML models, as measured by the consistency of their results across multiple training data cases. As shown in Figure 7, the spreads of the RC and ULNN models are the tightest in the general results, with their RMSE IQRs lying around 0.04°F . Additionally, although ULNN is a PIML model that does not explicitly incorporate the full physical equations, it still performs competitively with the standard RC model. In contrast, with limited data shown in Figure 8, the BLNN model demonstrates an outstanding result with the tightest spread of RMSEs, at approximately 0.01°F . Furthermore, all PIML methods consistently outperform the general NN model in terms of robustness, exhibiting significantly smaller RMSE spreads. Beyond the small training cases, with increased training data (Figure 9), these PAML models remain as robust as their PIML counterparts, except for BLNN. Even though BLNN shows slightly more uncertainty (a wider spread) with high training data access, its median RMSE always falls within the small-value range.

B. CASE STUDY: ORIGINAL DATA

Training and test data sets: This case contains 560 experimental data points in a series. The test data set consists of 30 points from index 530 to index 560, the training sets are selected from index 20 to index 300, with different sizes (in increments of 10, resulting in 29 cases). Therefore, the training data set is disjoint from the test data set. Here, the original data is combined with time-series data, so multiple predictions [81] and an RNN model will be considered in this scenario. Therefore, **thirteen morels** in Table 1 will be investigated in this section.

General comparison: Table 3 presents the RMSE for prediction across a wide range of training sizes in the original data case, from 20 to 200 training data points. In particular, the PAML models including LR, NN, SINDy, GP, ResNet, and NODE exhibit substantial variability and poor performance, especially at low training data sizes (from 20 to 40 training data points), with the median of RMSE larger than 0.8°F . On the other hand, the PIML models, especially BNN, ULNN, and BLNN, exhibit more stable, consistently lower RMSE across all training sizes than PAML models.

In this case, however, the RC model does not perform well, with RMSE typically above 0.7°F , due to an unknown disturbance from the fan that the model structure fails to capture. These findings underscore the importance of incorporating structural inductive biases, including boundedness and monotonicity, to enhance predictive robustness in realistic noise-affected scenarios.

Figures 11 and 12 present comparisons of the average RMSE for all models trained on the original data using 20 and 200 samples. At the low-data regime (Figure 11), PIML models, particularly BLNN and BSNN, significantly outperform PAML models, achieving RMSEs below 0.52°F . Conversely, SINDy and NN models exhibit poor performance, with RMSE as high as 1.40°F , reflecting their vulnerability to measurement noise and overfitting when trained on limited data. As the training size increases to 200 (Figure 12), the RMSE generally declines across the majority of models. However, PIML models, such as ULNN and BLNN, still provide the lowest RMSE of 0.49°F . These results strengthen the conclusion that integrating physical constraints and structural priors enhances model generalization, especially in practical conditions.

Comparison of PAML and PIML models: Generally speaking, the PIML models achieve higher accuracy than the PAML models (Table 3 and Figure 13). Among PAML methods, the GP model yields the best performance and is competitive with the PIML models, with RMSE frequently below 0.5°F when the training set exceeds 50 samples. By contrast, BLNN attains RMSE below 0.5°F even with as few as 20 training samples. The Table 3 also illustrates that BLNN seemingly achieves the smallest RMSEs in most cases of training data sizes. Figures 11 and 12 display the RMSE results for the original data with training data

TABLE 3. RMSE (°F) values of prediction across training data sizes in the original data case of thirteen models.

	Gray-box	Physics-agnostic ML							Physics-informed ML				
Size	RC	LR	NN	SINDy	GP	ResNet	NODE	RNN	BNN	MMNN	ULNN	BLNN	BSNN
20	0.672	0.796	0.910	1.397	0.490	0.803	0.800	0.619	0.630	0.795	0.672	0.439	0.516
30	0.731	0.830	0.808	0.838	0.903	0.834	0.830	0.638	0.611	0.830	0.561	0.423	0.516
40	0.738	0.854	0.827	0.859	0.895	0.854	0.859	0.649	0.563	0.854	0.503	0.408	0.879
50	0.658	0.680	0.520	0.742	0.559	0.681	0.682	0.561	0.499	0.680	0.426	0.383	0.678
60	0.649	0.680	0.546	1.208	0.416	0.680	0.681	0.675	0.515	0.544	0.439	0.434	0.689
70	0.675	0.707	0.695	1.228	0.404	0.707	0.708	0.618	0.512	0.514	0.483	0.509	0.716
80	0.695	0.710	0.563	1.216	0.373	0.711	0.713	0.624	0.495	0.531	0.419	0.408	0.715
90	0.687	0.713	0.524	1.237	0.351	0.719	0.714	0.626	0.511	0.713	0.445	0.415	0.719
100	0.712	0.729	0.534	0.530	0.413	0.729	0.730	0.630	0.508	0.480	0.441	0.429	0.405
110	0.728	0.741	0.521	0.580	0.536	0.750	0.743	0.639	0.531	0.496	0.452	0.424	0.747
120	0.736	0.733	0.565	0.625	0.577	0.734	0.735	0.626	0.532	0.496	0.461	0.435	0.417
130	0.761	0.757	0.552	0.744	0.545	0.759	0.759	0.646	0.543	0.757	0.472	0.448	0.757
140	0.782	0.780	0.581	0.788	0.507	0.781	0.781	0.668	0.557	0.502	0.503	0.507	0.780
150	0.788	0.790	0.608	0.812	0.427	0.790	0.790	0.672	0.557	0.521	0.493	0.486	0.790
160	0.766	0.780	0.648	0.681	0.519	0.780	0.780	0.668	0.554	0.544	0.509	0.535	0.780
170	0.762	0.774	0.604	0.667	0.505	0.774	0.774	0.666	0.546	0.542	0.497	0.486	0.774
180	0.766	0.777	0.619	0.694	0.486	0.777	0.777	0.668	0.536	0.777	0.488	0.481	0.769
190	0.767	0.776	0.610	0.701	0.425	0.776	0.777	0.667	0.534	0.776	0.490	0.481	0.770
200	0.775	0.777	0.687	0.720	0.508	0.777	0.778	0.668	0.548	0.777	0.491	0.487	0.777

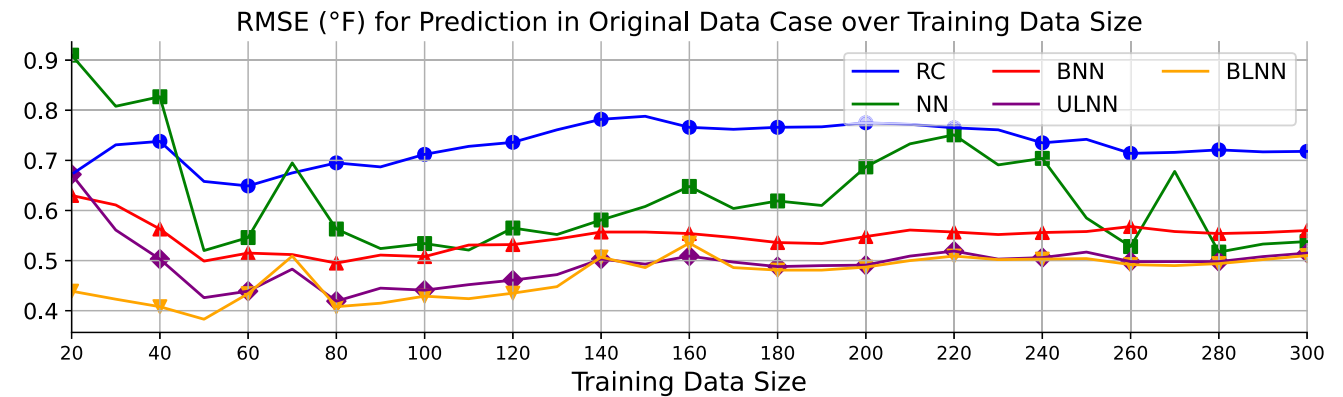


FIGURE 10. RMSE (°F) values of prediction across training data cases (from 20 to 300 points) in the original data case.

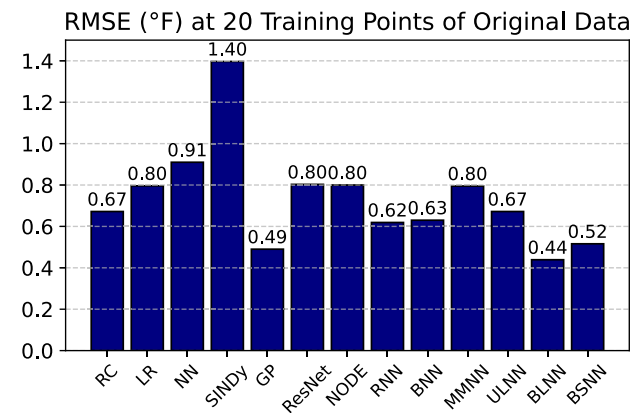


FIGURE 11. RMSE at 20 training data on the original set.

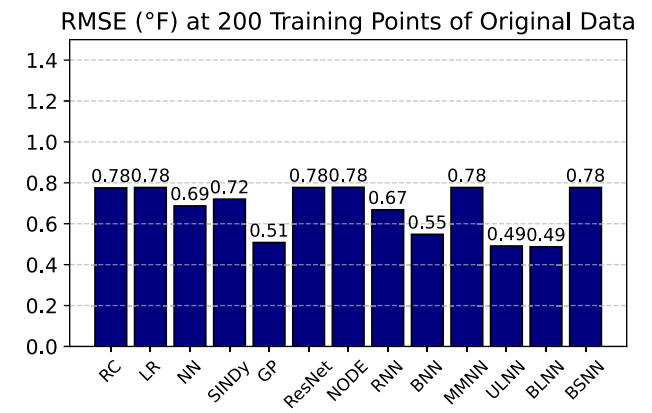


FIGURE 12. RMSE at 200 training data on the original set.

sizes of 20 and 200, respectively. Both the GP and BLNN models demonstrate high accuracy in identification, with RMSEs consistently below 0.5°F. Since there are unknown disturbances in the data, the RC model does not yield promising results as in the filtered data scenario, with

RMSEs of 0.5°F and 0.78°F when using training data sets of 20 and 200, respectively. Moreover, in Figure 11, the ResNet and NODE (with RMSE of 0.8°F) achieve better accuracy than the NN model. Specifically, the RNN and BNN with causality-based and boundedness-based improvements

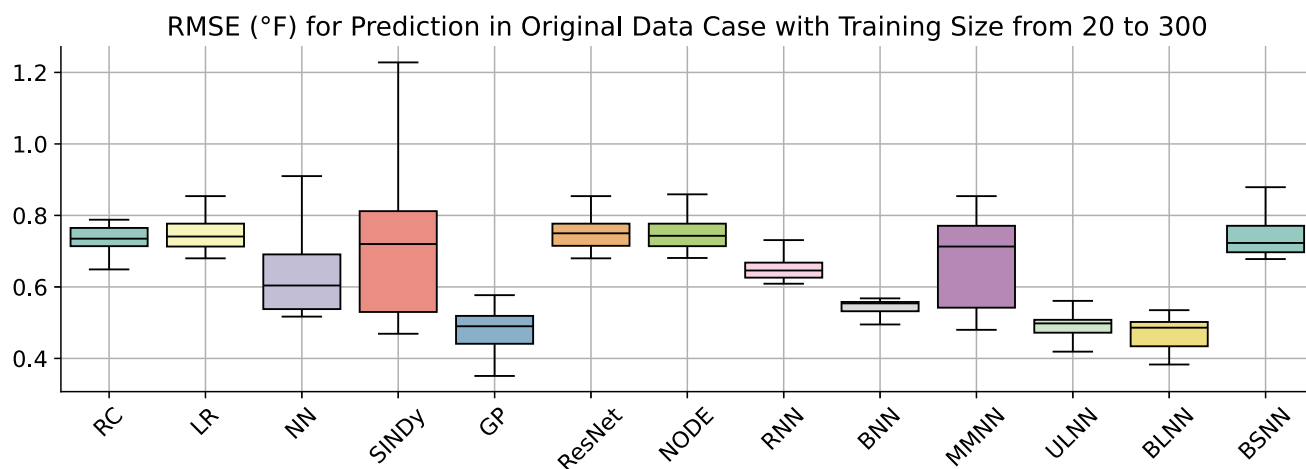


FIGURE 13. Box plot of RMSE (°F) for prediction performance across thirteen models in the original data case, evaluated with training sizes ranging from 20 to 300. Lower medians show higher accuracy, while narrower ranges reflect greater robustness.

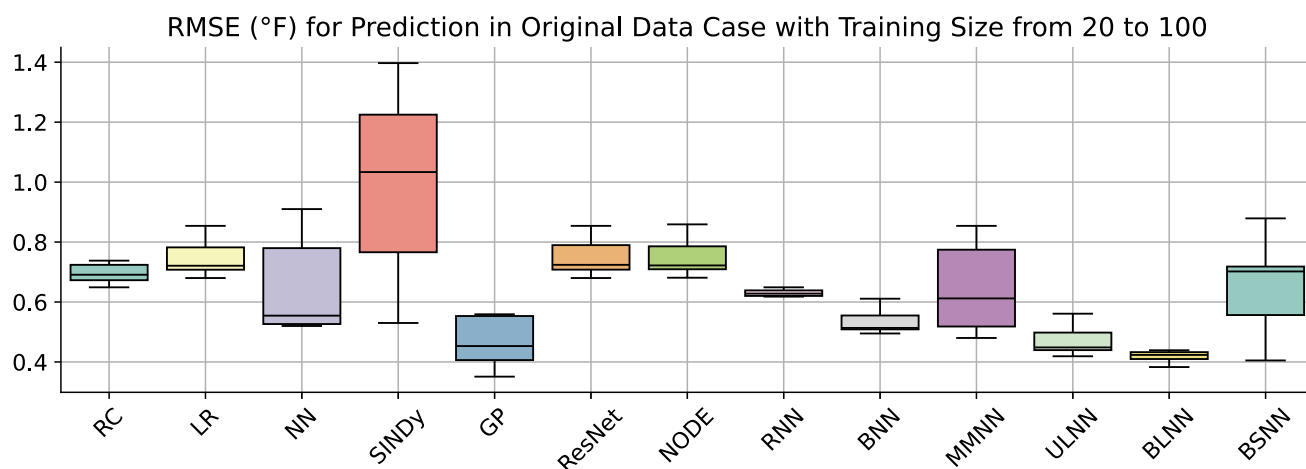


FIGURE 14. Box plot of RMSE (°F) for prediction performance across thirteen models in the original data case, evaluated with training sizes ranging from 20 to 100. Lower medians show higher accuracy, while narrower ranges reflect greater robustness.

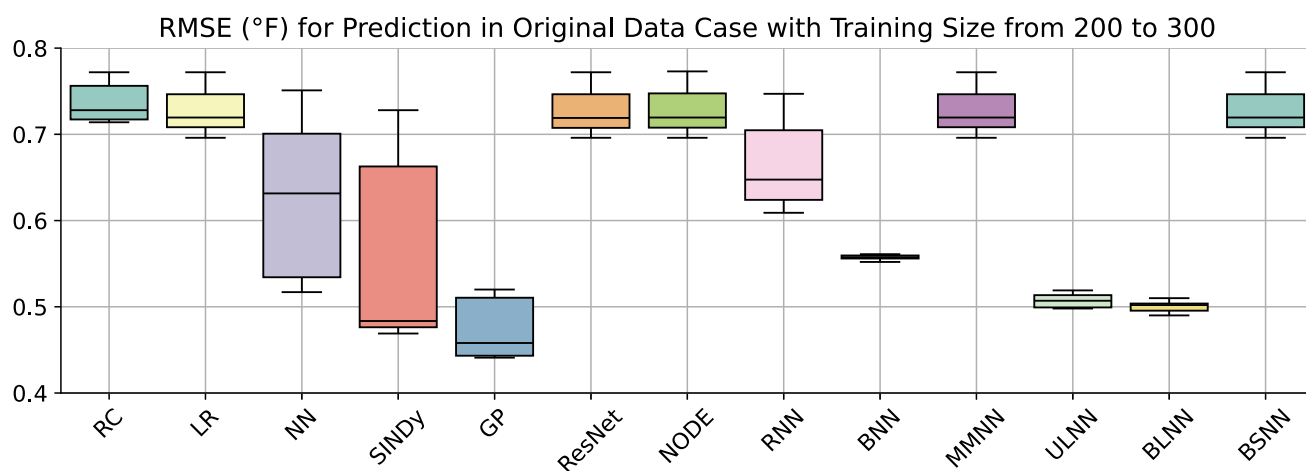


FIGURE 15. Box plot of RMSE (°F) for prediction performance across thirteen models in the original data case, evaluated with training sizes ranging from 200 to 300. Lower medians show higher accuracy, while narrower ranges reflect greater robustness.

enhance the NN from an RMSE of 0.91°F to 0.62°F and 0.63°F .

Figure 10 further shows that PIML models enforcing boundedness and monotonicity exhibit more stable and robust performance than the unconstrained NN model by achieving lower RMSE results (always smaller than 0.62°F). Likewise, BLNN and ULNN have similar performance when the training data size increases. Therefore, the boundedness-constraint contributes significantly to the precision of BLNN and a lower contribution than the monotonicity-constraint when the training data is large enough in the BLNN model. Moreover, in this original data set, although the NN model does not perform well as a PIML method, it still yields better results than the RC model. Thus, NN models, or ML models in general, are a good choice when working with systems that have significant unknown disturbances.

Statistical comparison: Figures 13, 14, and 15 present box plots of RMSE for all models trained on the original data case across three training size ranges: (i) from 20 to 300 data points (Figure 13), (ii) from 20 to 100 data points (Figure 14), and (iii) from 200 to 300 data points (Figure 15). In that matter, (i) represents the results of all 29 training data size scenarios, (ii) illustrates the performance of the models with low training data sizes, and (iii) shows the exhibition of thirteen models with sufficient training data sizes.

- **Accuracy:** In the low training data regime (Figure 14), the PIML models, especially ULNN, BLNN, and BNN, demonstrate significantly lower median errors of less than 0.6°F compared to traditional PAML models (except for the GP model). Here, the GP model also shows accurate performance, with a median around 0.5°F , similar to the ULNN and BLNN models, as GP can handle unknown noise during the training process. On the other hand, the SINDy model performs particularly poorly, with a median RMSE exceeding 1.0°F , highlighting that the terms used in SINDy do not adequately describe the unknown disturbances. The RC model performs reasonably well, with median results around 0.75°F , compared to the PAML methods, when the training data size is low (20-100 points). In contrast, investigating the case of large training data sizes (200-300 points; Figure 15), generally speaking, the median RMSE values for all 13 methods are lower than in the small training data cases of Figure 14. Therein, the GP model also demonstrates superior results, with the smallest median when given enough training data, at 0.45°F . Also, the BSNN model with the structure-informed machine learning approach does not achieve good accuracy compared to monotonicity-informed (ULNN, BLNN) and boundedness-informed (BNN, BLNN) models.
- **Robustness:** In terms of robustness, in the smaller 20–100 training range (Figure 14), both SINDy and NN demonstrate considerable volatility, evidenced by their large interquartile ranges (IQR around 0.4°F and 0.3°F , respectively) and whiskers. On the other hand,

models designed for structural constraints, namely RNN, BLNN, and ULNN, exhibit the highest robustness with extremely small IQRs ranging from about 0.01°F to 0.02°F ; the BLNN model, incorporating monotonicity and boundedness constraints, proves the most stable. Even with a moderate median RMSE (about 0.65°F), the RNN model's exceptional robustness makes it suitable for complex optimal controllers such as differentiable predictive control [49] or model predictive control [72]. Conversely, although the GP model has the best median accuracy (around 0.5°F), its relatively wide spread (IQR about 0.08°F) suggests potential reliability issues for long-horizon predictions. When the training size increases to the 200–300 range (Figure 15), the PAML models (BNN, ULNN, and BLNN) continue to dominate in robustness with IQRs of about 0.01°F , outperforming the RC model, which shows a slightly wider spread (IQR about 0.03°F). Furthermore, in this larger data regime, both ResNet and NODE appear more robust, with spreads of about 0.02°F , compared to the persistently wide spread of the NN model (IQR about 0.3°F).

C. DISCUSSIONS AND INSIGHTS

1) IMPACT OF DATA QUALITY

All ML models evaluated in this study are capable of accurately identifying the system. In particular, the RC model combines the RC structure with a data-driven optimization process for parameter estimation. Our experimental results indicate that the RC (gray-box) model is the most accurate model in the filtered data case. Even with a small training data set, it achieves accurate and robust system identification. Therefore, we recommend adopting gray-box modeling as an initial step for HVAC system identification, as it facilitates model interpretability and does not require extensive data collection from real systems.

Training models on the original data is more realistic than using the filtered data; therefore, results obtained from the original data should be emphasized, as they better represent experimental conditions. As discussed in Subsection VII-B, the Lattice NN, which embeds monotonicity constraints with two variants (BLNN and ULNN), achieves the best performance even with relatively small training data sets (fewer than 100 samples). Similarly, the BNN demonstrates promising results, showing improved accuracy and robustness compared to the standard NN model. Incorporating physical properties into ML models enhances their capability to identify system dynamics effectively. For systems influenced by unknown disturbances, ML models that integrate physical information such as boundedness, monotonicity, and structural constraints exhibit better generalization performance, resulting in smaller median errors, particularly under limited data conditions. Moreover, the RNN, which enforces causality across multi-step predictions, improves system robustness. Based on these findings, we suggest combining PIML models with RNN architectures to simultaneously enhance accuracy and robustness in HVAC system modeling.

2) IMPACT OF TRAINING DATA SIZE

We investigate the impact of limited training data, which is a common challenge when data collection in real systems is costly. With a minimal data set of only 20 training samples, the PIML models demonstrate their capability to accurately learn the HVAC system behavior, maintaining high accuracy and robustness as indicated by small RMSE and IQR values. When the data set features adequately capture the system dynamics, as in the filtered data case, the RC model, which is the most accurate and robust, requires only a small amount of training data. The RC model can also be regarded as a PIML approach because it integrates the optimization of a data-driven model within a parametric physical framework. Furthermore, as the training data size increases, the accuracy of all ML models improves progressively, approaching the performance achieved with the full training data set. Conversely, when the data is subject to noises and unknown disturbances, as in the original data case, and the available training data are limited, PIML techniques become favorable choices for HVAC modeling due to their superior learning accuracy under data-scarce conditions.

3) IMPACT OF PHYSICAL PROPERTY

Among the PIML models, the boundedness-constrained models (BNN and BLNN) and the RNN model exhibit the highest robustness in both the original and filtered data cases. The most accurate model is BLNN, which simultaneously incorporates monotonicity and boundedness constraints. These results suggest that, to improve model accuracy under limited training data, multiple physical properties should be integrated into the model architecture, introducing beneficial inductive biases. This conclusion motivates the design of the BSNN model, which incorporates three physical conditions to enhance the accuracy of system identification. Overall, the BNN, ULNN, and BLNN models outperform MMNN and BSNN in both accuracy and robustness across the original and filtered data sets. These findings indicate that boundedness and monotonicity provide essential physical insights into HVAC systems, enhancing the trustworthiness and interpretability of ML models.

Regarding the PAML methods, the GP model achieves the most accurate results. Moreover, when the GP model is trained with a sufficiently large data set (over 200 samples), its accuracy surpasses that of the BLNN model. We therefore suggest incorporating relevant physical properties of HVAC systems into the GP model to maintain accuracy under both limited and abundant training data. For example, [82] introduces monotonicity constraints in GP models, while the boundedness-constrained framework proposed in (20) for the BNN can similarly be extended to GP models, as demonstrated in [83]. However, the GP model may not be suitable for large-scale, high-dimensional systems due to the computational complexity associated with matrix inversion during training and prediction [32].

VIII. CONCLUSION

This paper presents a comprehensive comparative study of thirteen data-driven models for HVAC system identification, including seven PAML models, a gray-box model, and five PIML models with inductive biases, evaluated on real data. Our study shows that PIML models achieve superior accuracy in system identification compared to purely data-driven machine learning approaches. Among them, the multi-physics-informed model, such as the BLNN method, achieves the best performance in learning the real HVAC system across nearly all cases with varying training data sizes and different data sets (filtered and original data cases). On the other hand, the RC model remains a preferred choice for understanding the most important physical components in HVAC systems, as it requires less training data while still achieving reasonably accurate results. The results also highlight the potential of hybrid modeling strategies, such as integrating machine learning with RC models or combining multiple physical constraints within ML frameworks, to develop more robust and interpretable PIML models. This study provides a foundation for future work in designing advanced PIML approaches that leverage both data-driven learning and physical system knowledge for HVAC systems.

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